

KHALED GALAL YEHIA

Software Engineer — AI Solutions & Backend Development

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Summary

Software Engineer specializing in backend development, AI solutions, and system integration. Experienced in building enterprise-grade applications with Java/Spring Boot, Python, and TypeScript. Currently working in the R&D AI Solutions team at Intercom Enterprises, building AI-powered tools and automation solutions that support multiple teams across the company.

Experience

Intercom Enterprises

Cairo, Egypt

Software Engineer, R&D AI Solutions

01/2025 - present

Supported multiple teams across the company by building AI-powered tools and enterprise applications to automate and streamline their workflows.

- **QC Team:** Automated the QC workflow end-to-end with QC Agent (Flask + Angular 21 monorepo, Docker): it generates test case titles from an analysis document using Google Gemini 2.0 Flash and OpenAI GPT-4o with embedding-based de-duplication, elaborates approved titles into detailed steps, then executes each scenario against the real system and records a test video into a central library of all runs.
- **Company-wide:** Designed and developed the Correspondence Management System (CMS), an enterprise modular monolith using Spring Boot 4, Angular 21, PostgreSQL 16, and Flowable workflow engine.
- **Company-wide:** Built a production-ready RAG Agent using Flask, PostgreSQL with pgvector, and Google Generative AI, enabling employees to query company documents such as policies and internal guidelines through a conversational interface.
- **ECM Teams (FileNet & OpenText):** Automated the teams' documentation workflow — a full-stack generator (Angular 18 + Node.js/TypeScript + Google Gemini API) producing technical proposals, taxonomy definitions, and subscription event configurations, plus a Flask/LangChain FAISS-based RAG pipeline that drafts solution design documents, all from the same analysis documents.
- **Financial Team:** Built a bilingual (Arabic/English) document generator desktop application with Python and Tkinter to automate the generation of financial proposals.
- **Support Team:** Developed a multi-layer log filtering system in TypeScript to help first-layer support engineers investigate issues by isolating relevant entries from bank-submitted logs.

Intercom Enterprises

Cairo, Egypt

Back End Developer and Support

06/2024 - 12/2024

Supported multiple banking projects, ensuring the stability and performance of critical systems.

- Supported multiple banking projects including contributions to the United Bank, Banque du Caire, Arab International Bank, and Housing and Development Bank.
- Provided back-end development and support, ensuring the stability and performance of critical banking systems.

Intercom Enterprises

Cairo, Egypt

Back End Developer

10/2023 - 12/2023

Full-time position focusing on backend development for banking systems.

- Worked on the United Bank project, managing both front-end and back-end roles.
- Improved transaction speed by developing an instant payment process using Struts 2.

Intercom Enterprises

Cairo, Egypt

Digital Integration Trainee

08/2023 - 09/2023

Internship focusing on digital integration projects.

- Developed a fully functional CRUD demo for an online bookstore showcasing integration capabilities.
- Created an MVC demo focused on student data management.
- Integrated with OpenText for efficient document retrieval.

- Led the development of a Blood System API with IBM API Connect ensuring accurate ID validations.

Intercom Enterprises

Cairo, Egypt

RPA Developer Trainee

07/2022 - 09/2022

Trainee role focused on robotic process automation.

- Worked with automation technologies for robotic process automation tasks.
- Extracted data, filled forms, and moved files using RPA.

Intercom Enterprises

Cairo, Egypt

BPM Developer Trainee

02/2022 - 03/2022

Trainee role focused on business process management development.

- Car Mechanic Workflow: Designed and implemented a streamlined workflow in Appian BPM that efficiently checks and retrieves car data, automating the process of sending necessary information to mechanics for repairs.

Education

Quantic School of Business and Technology

Washington, DC (Online)

Master of Science in Software Engineering

06/2025 - 08/2026

DEAC-accredited program focused on AI-driven software engineering, system design, and modern development practices.

Ain Shams University

Cairo, Egypt

Bachelor of Computer and Information Science

09/2019 - 05/2023

Graduation Project

01/2023 - 06/2023

Driver Behavior Detection System — built in Python with MobileNet, VGG16 and VGG19 to detect and classify ten driver behaviours such as drowsiness and inattention, paired with a Flutter mobile application and a Flask backend for real-time monitoring and alerts.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C++

AI & LLM: LangChain, Retrieval-Augmented Generation (RAG), Google Gemini, OpenAI GPT-4o, FAISS, pgvector, vector embeddings, semantic de-duplication, prompt engineering

Back-End: Spring Boot, Flask, Node.js, REST API design and integration, Flowable workflow engine, Struts 2

Front-End: Angular, TypeScript, SCSS, HTML, CSS, Flutter

Databases: PostgreSQL, Elasticsearch, Redis, SQLite

Machine Learning: MobileNet, VGG16, VGG19, image classification, computer vision

DevOps & Tools: Docker, Docker Compose, Git, monorepo workspaces (npm/pnpm)

Enterprise & Automation: IBM API Connect, IBM FileNet, OpenText, Appian BPM, UiPath RPA

Certifications & Training

Digital Egypt Pioneers Program — Vulnerability Analyst / Penetration Tester — Ministry of Communications and Information Technology of Egypt (Infrastructure and Security). Nov 2024 - May 2025.

McKinsey.org Forward Program — Practical skills for problem-solving, communication, adaptability, and building a digital toolkit. Dec 2025.

Spring Boot 3 — Course focused on the Spring Boot framework to enhance backend development skills.

Comprehensive Cybersecurity Bootcamp — Intensive bootcamp covering various aspects of cybersecurity.

RPA Developer Foundation — Foundation course for RPA Development focused on automation technologies.

Appian Development — Course on Appian development covering business process management solutions.

Front End Diploma — Diploma covering front-end development technologies and frameworks.

Cybersecurity Fundamentals Workshop — Workshop introducing fundamental concepts in cybersecurity.